Prime extremality for product integrals over primitive sets

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Abstract

A set A of integers exceeding 1 is primitive if no element divides another. For $0 < c \le 1$ let $F_c(A) = \int_0^c \prod_{n\in A}(1-x^n)\,dx$. We prove that among primitive subsets of $\{2,\dots,N\}$ the primes up to N are, for every c, the unique minimizer of F_c, and that over all primitive sets the infimum is attained only by the set of all primes, although the primes do not minimize the integrand pointwise. The proof rests on one inequality, $\int_0^c \prod_{t\in T}(1-x^t)\,dx > \int_0^c (1-x)\,dx$ for every set $T \subseteq \{2,3,\dots\}$, which gives a strict compression replacing the multiples of an integer d by d. Consequently the divisibility-minimal elements of any set U uniquely minimize F_c among primitive subsets of U. The minimum at $c = 1$ is $\Lambda = 0.5506530112\ldots$. The truncation error satisfies $-\log(I_N - \Lambda) \sim 2\pi\sqrt{N/(3\log N)}$, with two further terms from the prime number theorem, and to within $O(\log N)$ it is fixed by the least prime exceeding N. The main results are formally verified in Lean 4.

Keywords. primitive sets; divisibility antichains; prime numbers; prime number theorem; partitions into primes; product inequalities; stochastic orders; formal verification.

1 Introduction

1.1 Primitive sets and product integrals

A set $A \subseteq \{2,3,\dots\}$ is *primitive* (a divisibility antichain) if no element of A divides another. Write $\mathcal{P}_N = \{p \le N : p \text{ prime}\}$ and $\mathcal{P}$ for the set of all primes. To a set $A \subseteq \{2,3,\dots\}$, finite or infinite, we attach the product

$$P_A(x) = \prod_{n\in A}(1-x^n) \qquad (0 \le x < 1),$$

which converges locally uniformly on $[0,1)$ (for infinite A because $\sum_{n\in A} x^n < \infty$), and its *initial integrals*

$$F_c(A) = \int_0^c P_A(x)\,dx, \qquad 0 < c \le 1.$$

For $A = \varnothing$ we set $P_\varnothing \equiv 1$, so $F_c(\varnothing) = c$.

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Why this functional? The product P_A is the reciprocal of the generating function of partitions with parts in A,

$$\frac{1}{P_A(x)} = \sum_{n\geq 0} p_A(n)\, x^n,$$

where $p_A(n)$ counts the partitions of n into parts from A, and its integral generalizes the classical evaluation of $\int_0^1 \prod_{n\geq 1}(1-x^n)\,dx$ through Euler's pentagonal theorem [16, 11]. We ask which primitive set of parts minimizes it. The answer is governed by a single comparison with the one-part product $1-x$ (Theorem 1.4 below).

Each P_A is nonincreasing in x and antitone in A: enlarging A lowers P_A pointwise. The extremal question is therefore not whether large primitive sets beat small ones but which *maximal* antichain the integrals prefer. The answer is the primes, uniformly in the truncation level c.

Theorem 1.1 (Unique finite minimizer)**.** *Let $N \geq 2$ and let $A \subseteq \{2,\dots,N\}$ be primitive with $A \neq \mathcal{P}_N$. Then*

$$F_c(\mathcal{P}_N) < F_c(A) \qquad \text{for every } c \in (0,1].$$

Theorem 1.2 (Infimum over all primitive sets; unique attainment)**.** *Let $Q(x) = \prod_p (1-x^p)$. For every $c \in (0,1]$,*

$$\inf_A F_c(A) = \int_0^c Q(x)\,dx =: \mathcal{L}(c),$$

the infimum over all *primitive sets $A \subseteq \{2,3,\dots\}$, finite or infinite, and it is attained precisely by $A = \mathcal{P}$: for every primitive $A \neq \mathcal{P}$ and every $c \in (0,1]$, $F_c(A) > \mathcal{L}(c)$. In particular no finite primitive set attains it, while the truncations $\mathcal{P}_N$ form a minimizing sequence. The value at $c = 1$ is*

$$\Lambda := \mathcal{L}(1) = \lim_{N\to\infty} \int_0^1 \prod_{p\leq N} (1-x^p)\,dx = 0.5506530112728\ldots$$

1.2 Pointwise extremality fails

The integrals are the right objects: the primes do *not* minimize the product itself. As $x \uparrow 1$ one has $P_A(x) \sim \big(\prod_{a\in A} a\big)(1-x)^{|A|}$ for finite A, so near 1 the product is governed by the cardinality of A, and maximal antichains in $\{2,\dots,N\}$ can have about $N/2$ elements, far more than $\pi(N)$. A concrete family already shows the failure at $N = 9$.

Proposition 1.3 (Failure of pointwise extremality)**.** *For $N \geq 9$ let $A_N = (\mathcal{P}_N \setminus \{2,3\}) \cup \{4,6,9\}$. Then A_N is primitive and*

$$\frac{P_{A_N}(x)}{P_{\mathcal{P}_N}(x)} = (1+x^2)(1+x^3)(1-x^9) \;\sim\; 36\,(1-x) \longrightarrow 0 \qquad (x \uparrow 1).$$

Hence $P_{A_N}(x) < P_{\mathcal{P}_N}(x)$ on a neighbourhood of 1, while $F_c(\mathcal{P}_N) < F_c(A_N)$ for every $c \in (0,1]$ by Theorem 1.1.

Proof. No prime ≥ 5 divides or is divisible by 4, 6 or 9, and these three are pairwise incomparable, so A_N is primitive. The ratio is $(1-x^4)(1-x^6)(1-x^9)/\big((1-x^2)(1-x^3)\big)$. □

The ratio drops below 1 at $x \approx 0.965$. Thus every initial integral prefers the primes although the integrand does not; the theorems identify an integrated comparison that survives a genuine obstruction to pointwise comparison (in probabilistic terms, the primes are extremal in increasing-concave order but not in the usual stochastic order; Remark 4.4).

1.3 The universal product inequality and prime compression

Both theorems are iterations of a single operation. If p is prime and the primitive set A contains multiples pt, $t \in T$, then $p \notin A$, and the *compression*

$$C_p(A) = \{p\} \cup \{a \in A : p \nmid a\}$$

is again primitive. Writing $B = \{a \in A : p \nmid a\}$, the change in the initial integral is

$$F_c(A) - F_c(C_p(A)) = \int_0^c P_B(u)\, E_T(u^p)\, du, \qquad E_T(v) = \prod_{t\in T}(1-v^t) - (1-v). \tag{1}$$

For $|T| \geq 2$ the factor E_T changes sign on $[0,1]$ (already $E_{\{2,3\}}(v) = v - v^2 - v^3 + v^5$), so the comparison cannot be pointwise. What is true is that its running integral is positive, uniformly over all sets of exponents.

Theorem 1.4 (Universal product inequality)**.** *For every nonempty set $T \subseteq \{2,3,\dots\}$, finite or infinite, and every $c \in (0,1]$,*

$$\int_0^c \prod_{t\in T}(1-v^t)\, dv \;>\; \int_0^c (1-v)\, dv = c - \frac{c^2}{2}.$$

Antitonicity reduces the theorem to $T = \{2,3,\dots\}$, where $\prod_{n\geq 2}(1-x^n) - (1-x)$ changes sign exactly once, so only the endpoint $c = 1$ must be checked (Section 2). The set hypothesis is essential (Remark 2.3). Section 3 transports the inequality through the substitution $v = u^p$ and the weight P_B in (1), giving the strict compression $F_c(C_p(A)) < F_c(A)$, and Section 4 iterates it.

1.4 Extremality relative to an ambient set

Primality of p plays no role in the compression inequality: the substitution $v = u^d$ works for every integer $d \geq 1$. What primality secures is that $C_p(A)$ is again primitive. Replacing the primes by arbitrary divisors gives a relative theorem. For $U \subseteq \{2,3,\dots\}$ let

$$\mathrm{Min}(U) = \{u \in U : \text{ no proper divisor of } u \text{ lies in } U\}$$

be the set of divisibility-minimal elements of U. It is primitive, and Theorem 4.1 states that among primitive subsets of U it is, for every $c \in (0,1]$, the unique minimizer of F_c, for finite and infinite sets alike. The case $U = \{2,3,\dots\}$ recovers Theorem 1.2. Two other cases correspond to recent work on the additive weight $f(A) = \sum_{a\in A} 1/(a\log a)$ (Corollary 4.2). For primitive sets contained in $[x,\infty)$, the problem of Erdős, Sárközy and Szemerédi that Alexeev et al. [1] settle asymptotically, $f(A) \leq 1 + O(1/\log x)$, has for F_c an exact unique extremal set, the integers $n \geq x$ whose largest proper divisor is less than x. For primitive sets of integers with at least k prime factors, all taken from a set of primes $\mathcal{Q}$, the minimizer is the set of integers with exactly k such factors. This is the analogue of a conjecture of Banks and Martin [3], and it holds for every $\mathcal{Q}$, whereas the additive statement fails when $2 \in \mathcal{Q}$ [1].

1.5 The truncations and the prime number theorem

The minimizing sequence of Theorem 1.2 is $I_N = F_1(\mathcal{P}_N) \downarrow \Lambda$. How fast it converges is a question about the primes beyond N, and the answer is governed by their distribution.

Theorem 1.5 (Tail asymptotics)**.** *As $N \to \infty$,*

$$-\log\bigl(I_N - \Lambda\bigr) \sim 2\pi\sqrt{\frac{N}{3\log N}}.$$

Consequently, if p_n is the nth prime and $J_n = I_{p_n}$, then $-\log(J_n - \Lambda) \sim (2\pi/\sqrt{3})\sqrt{n}$.

With $\Phi(t) = -\log\prod_p(1 - e^{-pt})$ and $M_N = \inf_{t>0}(Nt + \Phi(t))$, the bound $I_N - \Lambda \le eN^2e^{-M_N}$ is elementary and holds for every $N \ge 3$. The prime number theorem is needed exactly to evaluate M_N and to place a prime in $(N, (1+\varepsilon)N]$ (Section 5.2); Chebyshev's bounds give only the order of magnitude (Remark 5.4). The leading term is also the logarithmic growth rate of the number of partitions of N into primes, because both come from the same saddle point (Remark 5.11).

Since I_N changes only when N passes a prime, the natural variable is the least prime $q_N > N$. Elementary estimates show $\bigl|-\log(I_N - \Lambda) - M_{q_N}\bigr| \le 2\log q_N + 4$ (Proposition 5.5). Consequently, the bound $-\log(I_N - \Lambda) \le M_N + O(\log N)$ is equivalent to the prime-gap estimate $p_{n+1} - p_n \ll \sqrt{p_n}(\log p_n)^{3/2}$ (Corollary 5.6). That estimate follows from the Riemann hypothesis but is not known unconditionally.

Relative lower-order terms, by contrast, are available unconditionally. The prime number theorem with a classical error term gives

$$-\log(I_N - \Lambda) = 2\pi\sqrt{\frac{N}{3\ell}}\Bigl(1 - \frac{\log\ell}{2\ell} + \frac{C}{\ell} + O\Bigl(\frac{(\log\ell)^2}{\ell^2}\Bigr)\Bigr), \qquad \ell = \log N,$$

with $C = \gamma - \zeta'(2)/\zeta(2) + \frac{1}{2}\log(2\zeta(2)) = 1.7426\ldots$ (Theorem 5.10). Its remainder is much larger than $\log N$, so this is consistent with Corollary 5.6.

1.6 Context

For the additive weight $\sum_{a\in A} 1/(a\log a)$, extremality of the primes over primitive sets is the Erdős conjecture, proved by Lichtman [8]; see Lichtman–Pomerance [9] for the history and [1] for recent developments. The present paper concerns a different, multiplicative functional on the same class of sets, with an elementary proof; it is not a new route to the additive problem. The comparison is nonetheless instructive: for linear weights $\sum_{a\in A} w(a)$ the primes win only with sufficient damping (for $w(a) = 1/a$ they lose: the integers in $[2, N]$ with exactly k prime factors, $k \approx \log\log N$, form a primitive set whose reciprocal sum is of order $\log N/\sqrt{\log\log N}$; see [9] and the references there), whereas the product integrals prefer the primes for every truncation level, with strict uniqueness and no dependence on N, by a mechanism that is a single elementary inequality.

Integrals of this shape have precedents. The all-exponent member $\int_0^1 \prod_{n\ge1}(1 - x^n)\,dx$ is Sandham's Monthly Problem 4614 [16], evaluated in closed form by Mulcrone [11] through Euler's pentagonal theorem (OEIS A258232); the even member is OEIS A258408. For the prime member no closed form appears to be recorded. Its decimal expansion is OEIS A395518,

contributed by the author, whose entry states the finite-antichain infimum at $c = 1$; the present paper supplies the proofs, the uniformity in c, the strict uniqueness, and the infinite case.

2 The universal product inequality

Throughout, $H(x) = \prod_{n\geq 2}(1 - x^n)$ for $0 \leq x < 1$, and $H(1) = 0$. The product converges locally uniformly on $[0, 1)$, and $0 \leq H(x) \leq 1 - x^2 \to 0$ as $x \uparrow 1$, so H is continuous on $[0, 1]$. For every set $T \subseteq \{2, 3, \dots\}$ and $x \in [0, 1]$,

$$P_T(x) \;\geq\; H(x), \tag{2}$$

since every factor omitted from P_T lies in $[0, 1]$. Two lemmas control the function $H(x) - (1 - x)$.

Lemma 2.1 (Single crossing)**.** *There is exactly one $x^* \in (0, 1)$ with $H(x^*) = 1 - x^*$, and $H(x) - (1 - x)$ is strictly positive on $(0, x^*)$ and strictly negative on $(x^*, 1)$.*

Proof. On $[0, 1)$ write

$$h(x) = \log \frac{H(x)}{1 - x} = \log(1 + x) + \sum_{n\geq 3} \log(1 - x^n),$$

using $(1 - x^2)/(1 - x) = 1 + x$. The series converges, together with the series of first and second derivatives, locally uniformly on $[0, 1)$ (each is dominated by a multiple of $\sum n^2 x^{n-2}$ on $[0, r]$, $r < 1$), so h is twice differentiable there with

$$h''(x) = -\frac{1}{(1 + x)^2} - \sum_{n\geq 3} \frac{n\, x^{n-2}\big((n - 1) + x^n\big)}{(1 - x^n)^2} \;<\; 0.$$

Thus h is strictly concave on $[0, 1)$. Moreover $h(0) = 0$, $h'(0) = 1 > 0$, and $h(x) \leq \log(1 + x) + \log(1 - x^3) \to -\infty$ as $x \uparrow 1$. A strictly concave function that vanishes at 0 and increases there is positive on some interval $(0, \delta)$, and the set where it is nonnegative is an interval; since $h \to -\infty$, that interval is $[0, x^*]$ for a unique $x^* \in (0, 1)$, with $h > 0$ on $(0, x^*)$ and $h < 0$ on $(x^*, 1)$. Since $H(x) - (1 - x) = (1 - x)(e^{h(x)} - 1)$ has the sign of h on $(0, 1)$, the lemma follows. □

Lemma 2.2 (Endpoint estimate)**.**

$$\int_0^1 H(x)\, dx - \frac{1}{2} \;\geq\; \frac{34619}{4476780} \;>\; 0.$$

Proof. For numbers $y_n \in [0, 1]$ one has $\prod(1 - y_n) \geq 1 - \sum y_n$ (the inequality is trivial when the right side is negative and follows by induction otherwise). Applying this to the factors with $n \geq 6$ and keeping the factors $2, \dots, 5$ exactly gives, for $0 \leq x < 1$,

$$H(x) \;\geq\; \prod_{n=2}^{5}(1 - x^n)\Big(1 - \frac{x^6}{1 - x}\Big) = (1 + x)(1 - x^3)(1 - x^4)(1 - x^5)\,(1 - x - x^6),$$

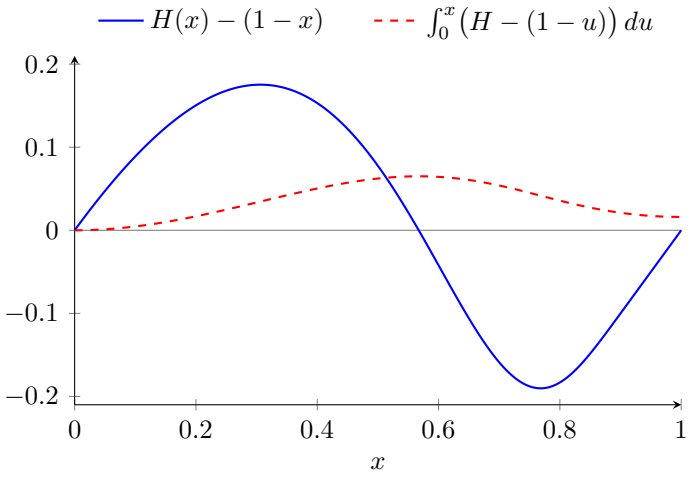


Figure 1: The extremal case $T = \{2, 3, \dots\}$ of Theorem 1.4. The integrand $H(x) - (1-x)$ crosses zero exactly once, at $x^* \approx 0.567$, so its running integral rises and then falls to the endpoint value $\int_0^1 H - \frac{1}{2} \approx 0.0161$, which is positive.

a polynomial of degree 19 (the bound may be negative near 1, which does no harm). Exact integration gives

$$\int_0^1 \prod_{n=2}^{5} (1 - x^n)\,dx = \frac{96601}{180180}, \qquad \int_0^1 \frac{x^6 \prod_{n=2}^{5}(1-x^n)}{1-x}\,dx = \frac{275501}{9699690},$$

using $\prod_{n=2}^{5}(1-x^n) = 1 - x^2 - x^3 - x^4 + x^6 + 2x^7 + x^8 - x^{10} - x^{11} - x^{12} + x^{14}$ and $(1+x)(1-x^3)(1-x^4)(1-x^5) = 1 + x - x^3 - 2x^4 - 2x^5 - x^6 + x^7 + 2x^8 + 2x^9 + x^{10} - x^{12} - x^{13}$. The difference of the two integrals minus $\frac{1}{2}$ is $34619/4476780$. □

The true value is $\int_0^1 H\,dx - \frac{1}{2} = 0.01609\ldots$, so the estimate loses only a factor of about two; the crossing is at $x^* = 0.567\ldots$ (Figure 1).

Proof of Theorem 1.4. Let $R(c) = \int_0^c \big(H(x) - (1-x)\big)dx$. By Lemma 2.1, R is strictly increasing on $[0, x^*]$ and strictly decreasing on $[x^*, 1]$, with $R(0) = 0$. Hence $R(c) > 0$ for $c \in (0, x^*]$, and for $c \in [x^*, 1]$, $R(c) \geq R(1) > 0$ by Lemma 2.2. So $R > 0$ on $(0, 1]$. By (2), $\int_0^c P_T \geq \int_0^c H$ for every T, and the theorem follows. □

Remark 2.3 (Sets versus multisets). The function H is the pointwise infimum of P_T over *sets* T, and that is exactly what (2) uses. For multisets the inequality fails: with three copies of the exponent 2, $\int_0^1 (1-v^2)^3\,dv = \frac{16}{35} < \frac{1}{2}$, and indeed $\int_0^c (1-v^2)^3\,dv < c - c^2/2$ for all $c > 0.6451\ldots$. Multisets never arise below because A is a set.

3 Transport and strict compression

The block exchange (1) needs Theorem 1.4 after the substitution $v = u^p$ and under a nonincreasing weight P_B. The lemmas below transport the estimate, and Proposition 3.4 combines them.

Lemma 3.1 (Singular monotone weights)**.** *Let g be continuous on $[0,1]$ with $G(y) = \int_0^y g > 0$ for all $y \in (0,1]$, and let w be positive, nonincreasing and C^1 on $(0,1]$, integrable at 0, with $w(\varepsilon)G(\varepsilon) \to 0$ as $\varepsilon \downarrow 0$. Then for every $c \in (0,1]$,*

$$\int_0^c w(v)g(v)\,dv \;\geq\; w(c)\,G(c) \;>\; 0.$$

Proof. Integrate by parts on $[\varepsilon, c]$: $\int_\varepsilon^c wg = w(c)G(c) - w(\varepsilon)G(\varepsilon) - \int_\varepsilon^c w'G$. The last integral is nonpositive since $w' \leq 0$ and $G \geq 0$, and $w(\varepsilon)G(\varepsilon) \to 0$; let $\varepsilon \downarrow 0$ (the left side converges since wg is integrable at 0). □

Corollary 3.2 (Power substitution)**.** *Let $p \geq 1$ be an integer and let $T \subseteq \{2,3,\dots\}$ be nonempty, finite or infinite. Then for every $c \in (0,1]$,*

$$\int_0^c E_T(u^p)\,du > 0.$$

Proof. Let $G(y) = \int_0^y E_T(v)\,dv$, so $G > 0$ on $(0,1]$ by Theorem 1.4, and $G(y) = O(y)$ at 0 since E_T is continuous on $[0,1]$. Substituting $v = u^p$,

$$\int_0^c E_T(u^p)\,du = \frac{1}{p}\int_0^{c^p} v^{1/p-1}E_T(v)\,dv,$$

and $w(v) = v^{1/p-1}/p$ is positive, nonincreasing, integrable at 0, with $w(\varepsilon)G(\varepsilon) = O(\varepsilon^{1/p}) \to 0$; apply Lemma 3.1. □

Lemma 3.3 (Monotone weights)**.** *Let g be continuous on $[0,1]$ with $G(y) = \int_0^y g \geq 0$ for all $y \in [0,1]$, and let w be nonnegative, continuous and nonincreasing on $[0,1]$. Then for every $c \in (0,1]$,*

$$\int_0^c w(u)g(u)\,du = w(c)G(c) + \int_{(0,c]} G\,d(-w) \;\geq\; 0,$$

where $d(-w)$ is the finite nonnegative Borel measure on $(0,c]$ with $d(-w)\big((a,b]\big) = w(a) - w(b)$. The left side is strictly positive if $G > 0$ on $(0,c]$ and either $w(c) > 0$ or $d(-w)$ has positive mass on $(0,c)$.

Proof. Since w is continuous and nonincreasing, $d(-w)$ is a finite nonnegative measure without atoms, of total mass $w(0) - w(c)$. As $G' = g$ is continuous, $\int_0^c wg\,du = \int_0^c w\,dG$, and Riemann–Stieltjes integration by parts with $G(0) = 0$ gives the stated identity; both terms are nonnegative. For strictness: if $w(c) > 0$ then $w(c)G(c) > 0$; if $d(-w)$ has positive mass on $(0,c)$, then, G being continuous and strictly positive there, $\int G\,d(-w) > 0$. □

The Stieltjes form is needed because the weight will be an infinite product P_B, which is continuous and nonincreasing on $[0,1]$ but need not be C^1 at the endpoint; Lemma 3.1, by contrast, allows a singularity at 0 but needs a smooth weight.

Proposition 3.4 (Strict block exchange)**.** *Let $p \geq 1$ be an integer (primality is not needed) and let $B, T \subseteq \{2,3,\dots\}$ be finite or infinite, T nonempty, with B disjoint from $pT = \{pt : t \in T\}$ and $p \notin B$. Set $A = B \cup pT$ and $A' = B \cup \{p\}$. Then for every $c \in (0,1]$,*

$$F_c(A') < F_c(A).$$

Proof. Pointwise, $P_A(u) - P_{A'}(u) = P_B(u)\big(\prod_{t\in T}(1-u^{pt}) - (1-u^p)\big) = P_B(u)E_T(u^p)$, so

$$F_c(A) - F_c(A') = \int_0^c P_B(u)\, E_T(u^p)\, du.$$

The weight $w = P_B$ is nonnegative, continuous and nonincreasing on $[0,1]$, and $G(y) = \int_0^y E_T(u^p)\, du > 0$ on $(0,1]$ by Corollary 3.2. If $c < 1$ then $P_B(c) > 0$ and Lemma 3.3 gives strict positivity. If $c = 1$: either $B = \varnothing$, and the integral is $\int_0^1 E_T(u^p)\, du > 0$ directly; or $B \neq \varnothing$, in which case $P_B(1) = 0$ and $d(-P_B)$ has total mass $P_B(0) - P_B(1) = 1$, carried by $(0,1)$ since P_B is continuous, while $G > 0$ there; the strictness clause of Lemma 3.3 applies. □

4 Proofs of the extremality theorems

For a primitive A containing a composite m and a prime $p \mid m$, primitivity gives $p \notin A$, and the compression $C_p(A) = \{p\} \cup \{a \in A : p \nmid a\}$ is a block exchange in the sense of Proposition 3.4 with $B = \{a \in A : p \nmid a\}$ and nonempty $T = \{a/p : a \in A,\ p \mid a\}$. It preserves primitivity: inherited nonmultiples remain incomparable, p divides none of them, and none divides p.

Proof of Theorem 1.1. Let $A \subseteq \{2,\dots,N\}$ be primitive, $A \neq \mathcal{P}_N$, and fix $c \in (0,1]$. If A contains a composite m, choose a prime $p \mid m$; then $F_c(C_p(A)) < F_c(A)$ by Proposition 3.4. The compression strictly lowers the element sum, since a nonempty block of multiples of p, all at least $2p$, is replaced by p. Induction on the element sum terminates at an all-prime set $R \subseteq \mathcal{P}_N$ with $F_c(R) < F_c(A)$, strict inequality being secured by the first compression step.

If instead A contains no composite, then $A \subseteq \mathcal{P}_N$ with $A \neq \mathcal{P}_N$, and we take $R = A$. In either case, if $R \neq \mathcal{P}_N$, add the missing primes one at a time: for $q \in \mathcal{P}_N \setminus R$,

$$F_c(R \cup \{q\}) = F_c(R) - \int_0^c x^q P_R(x)\, dx < F_c(R),$$

since the integrand is strictly positive on $(0,c)$. Chaining, $F_c(\mathcal{P}_N) < F_c(A)$. □

Proof of Theorem 1.2. Fix $c \in (0,1]$.

Lower bound. For finite primitive $A \subseteq \{2,\dots,N\}$, Theorem 1.1 (with its trivial case $A = \mathcal{P}_N$) gives $F_c(A) \geq F_c(\mathcal{P}_N)$. The integrands $P_{\mathcal{P}_N}$ decrease pointwise to Q and are bounded by 1, so by dominated convergence $F_c(\mathcal{P}_N) \downarrow \int_0^c Q = \mathcal{L}(c)$; hence $F_c(A) \geq \mathcal{L}(c)$ for every finite primitive A. For infinite primitive A, the truncations $A \cap [2,N]$ are finite primitive and $F_c(A) = \lim_N F_c(A \cap [2,N]) \geq \mathcal{L}(c)$, again by dominated convergence. Attainment at the primes is the definition: $F_c(\mathcal{P}) = \int_0^c Q = \mathcal{L}(c)$.

Uniqueness. Let $A \neq \mathcal{P}$ be primitive; we show $F_c(A) > \mathcal{L}(c)$. If $A = \varnothing$ then $F_c(A) = c > \mathcal{L}(c)$ since $Q < 1$ on $(0,c)$. If A contains a composite m, choose a prime $p \mid m$; the compression $C_p(A)$ is a block exchange with nonempty T (possibly infinite T and B), preserves primitivity, and by Proposition 3.4 satisfies $F_c(C_p(A)) < F_c(A)$. Therefore $F_c(A) > F_c(C_p(A)) \geq \mathcal{L}(c)$, the second inequality being the lower bound applied to the primitive set $C_p(A)$. If A contains no composite, then $A \subsetneq \mathcal{P}$; choose a prime $q \notin A$. Then $A \cup \{q\}$ is primitive and

$$F_c(A) - F_c(A \cup \{q\}) = \int_0^c x^q P_A(x)\, dx > 0,$$

since $P_A > 0$ on $(0,1)$, so $F_c(A) > F_c(A \cup \{q\}) \geq \mathcal{L}(c)$.

Non-attainment by finite sets. Every finite primitive set differs from $\mathcal{P}$, so the strict inequality above proves non-attainment; the minimizing sequence $\mathcal{P}_N$ was exhibited in the lower bound. The decimal expansion of $\Lambda = \mathcal{L}(1)$ is certified in Section 5.1. □

4.1 Extremality relative to an ambient set

For $U \subseteq \{2, 3, \dots\}$ let $\mathrm{Min}(U)$ be the set of $u \in U$ with no proper divisor in U. It is primitive: if $a \mid b$ with $a \neq b$ both in $\mathrm{Min}(U)$, then a is a proper divisor of b lying in U.

Theorem 4.1 (Relative extremality)**.** *Let $U \subseteq \{2, 3, \dots\}$ and let $A \subseteq U$ be primitive, finite or infinite, with $A \neq \mathrm{Min}(U)$. Then*

$$F_c(\mathrm{Min}(U)) < F_c(A) \qquad \text{for every } c \in (0, 1].$$

Likewise, for every N and every primitive $A \subseteq U \cap [2, N]$ with $A \neq \mathrm{Min}(U) \cap [2, N]$, $F_c(\mathrm{Min}(U) \cap [2, N]) < F_c(A)$ for every $c \in (0, 1]$.

Proof. Let $A \subseteq U$ be primitive and let $a \in A \setminus \mathrm{Min}(U)$. Choose a proper divisor $d \in U$ of a. No element $b \in A$ divides d: otherwise $b \mid a$ and so $b = a$ by primitivity, whence $a \mid d \mid a$ and $d = a$. In particular $d \notin A$. Hence $C_d(A) = \{d\} \cup \{b \in A : d \nmid b\}$ is primitive, as in Section 4, and $C_d(A) \subseteq U$. Moreover $F_c(C_d(A)) < F_c(A)$ by Proposition 3.4, applied with $p = d$, $B = \{b \in A : d \nmid b\}$ and $T = \{b/d : b \in A,\ d \mid b\} \ni a/d$. Every element of T is at least 2 because $d \notin A$.

The proofs of Theorems 1.1 and 1.2 now go through verbatim, with the compressions C_d in place of C_p and $\mathrm{Min}(U)$ in place of the primes. In the finite case, compression strictly lowers the element sum, and a set contained in $\mathrm{Min}(U) \cap [2, N]$ is completed by adding the missing elements one at a time, each addition strictly lowering F_c. For the lower bound in the infinite case, the truncations $A \cap [2, N]$ are primitive subsets of $U \cap [2, N]$, and $\mathrm{Min}(U \cap [2, N]) = \mathrm{Min}(U) \cap [2, N]$ because divisors are smaller than their multiples; dominated convergence then gives $F_c(\mathrm{Min}(U)) \leq F_c(A)$. For strictness, either $A \subsetneq \mathrm{Min}(U)$, and adding a missing element of $\mathrm{Min}(U)$ keeps the set primitive and strictly lowers F_c; or A has an element outside $\mathrm{Min}(U)$, and one compression C_d strictly lowers F_c. In both cases the lower bound, applied to the new set, finishes the proof. □

For $U = \{2, 3, \dots\}$, $\mathrm{Min}(U)$ is the set of primes, and Theorem 4.1 contains Theorems 1.1 and 1.2. Write $\Omega(n)$ for the number of prime factors of n counted with multiplicity and $P^-(n)$ for the least prime factor.

Corollary 4.2. *For every $c \in (0, 1]$ the following sets are the unique minimizers of F_c in the stated classes.*

(a) Large numbers. *Among primitive sets contained in $[x, \infty)$, $x \geq 2$: the set $S_x = \{n \geq x : n/P^-(n) < x\}$ of integers $n \geq x$ whose largest proper divisor is less than x. It contains $[x, 2x)$ and every prime $\geq x$, and $S_2 = \mathcal{P}$.*

(b) Restricted primes. *Let $\mathcal{Q}$ be a nonempty set of primes, let $\mathbb{N}_k(\mathcal{Q})$ be the set of integers with $\Omega(n) = k$ all of whose prime factors lie in $\mathcal{Q}$, and $\mathbb{N}_{\geq k}(\mathcal{Q}) = \bigcup_{j \geq k} \mathbb{N}_j(\mathcal{Q})$. Among primitive subsets of $\mathbb{N}_{\geq k}(\mathcal{Q})$: the set $\mathbb{N}_k(\mathcal{Q})$. In particular*

$$F_c(\mathbb{N}_1(\mathcal{Q})) < F_c(\mathbb{N}_2(\mathcal{Q})) < F_c(\mathbb{N}_3(\mathcal{Q})) < \cdots .$$

(c) Hilbert numbers. *Among primitive sets of integers $n > 1$ with $n \equiv 1 \pmod 4$: the irreducible elements of the multiplicative monoid $1 + 4\mathbb{N}$ (for instance $5, 9, 13, 21, 49$).*

Proof. In each case we identify $\mathrm{Min}(U)$ and apply Theorem 4.1. (a) A proper divisor of n lies in $[x, \infty)$ if and only if the largest one, $n/P^-(n)$, is at least x. For $n < 2x$ it is at most $n/2 < x$. (b) If $\Omega(n) > k$, deleting prime factors of n produces a proper divisor in $\mathbb{N}_k(\mathcal{Q})$; if $\Omega(n) = k$, every proper divisor has fewer than k prime factors. The chain follows because $\mathbb{N}_{k+1}(\mathcal{Q})$ is a primitive subset of $\mathbb{N}_{\ge k}(\mathcal{Q})$ different from the nonempty set $\mathbb{N}_k(\mathcal{Q})$. (c) This is the definition of irreducibility in the monoid. □

Remark 4.3 (Comparison with the additive weight). For $f(A) = \sum_{a \in A} 1/(a \log a)$, the problem of Corollary 4.2(a) is a 1966 question of Erdős, Sárközy and Szemerédi. Alexeev et al. [1] prove $f(A) \le 1 + O(1/\log x)$ for primitive $A \subseteq [x, \infty)$, by a Markov chain with von Mangoldt weights that moves from n to n/q. The analogue of (b) for f is a conjecture of Banks and Martin [3]; it is false when $2 \in \mathcal{Q}$ and is proved in [1] for sets of odd primes. For the product integrals both statements are exact and hold without restriction. The reason is that the compression inequality behind Theorem 4.1 holds uniformly in the divisor d and in the block of multiples dT, while linear weights must balance the lost multiples against the single gained divisor.

Remark 4.4 (Stochastic order and moments). Let M_n be the maximum of n independent uniform $[0, 1]$ variables, the families independent as n varies, and let $Z_A = \inf_{n \in A} M_n$ ($Z_\varnothing = 1$). By Borel–Cantelli the infimum is attained almost surely for infinite A, so $\mathbb{P}(Z_A > x) = P_A(x)$ and $F_c(A) = \mathbb{E}\min(Z_A, c)$. The hinge functions $\min(\,\cdot\,, c)$ generate the increasing-concave order $\preceq_{\mathrm{icv}}$ [17]: after replacing $\phi(0)$ by $\phi(0^+)$, which changes no expectation since $Z_A > 0$ almost surely, every increasing concave ϕ on $[0, 1]$ has the form $\phi(0^+) + \alpha x + \int_{(0,1]} \min(x, t)\, \mu(dt)$ with $\alpha \ge 0$ and $\mu \ge 0$. Theorems 1.1 and 1.2 therefore give $Z_{\mathcal{P}_N} \preceq_{\mathrm{icv}} Z_A$ for primitive $A \subseteq \{2, \dots, N\}$ and $Z_{\mathcal{P}} \preceq_{\mathrm{icv}} Z_A$ for every primitive A, whereas by Proposition 1.3 the usual stochastic order fails.

The extremality also reaches beyond concave test functions: for $0 < k \le 17/4$ the primes uniquely minimize $\mathbb{E}\min(Z_A^k, c^k) = \int_0^c k u^{k-1} P_A(u)\, du$, for finite and infinite primitive sets alike; for instance $x + \varepsilon x^2$ singles out the primes for every $\varepsilon > 0$. A complete proof is formalized (Section 6); we sketch it. For $k \le 1$ it follows from the case $k = 1$ by integration by parts. For $k > 1$ one repeats Sections 3 and 4 with the weight $k u^{k-1}$. A compression at $p \ge k$ is handled by Lemma 3.1. For $p < k$, put $s = k/p$ and $J_s(z) = \int_0^z x^{s-1}\big(Q(x) - (1 - x)\big)dx$; the change in the integral splits into the corresponding comparison of the quotient block $T = \{a/p\}$ with the primes at exponent s, which is nonnegative by induction on k because T is itself primitive, plus $J_s(y^p)$. As in Lemma 2.1, $Q(x) - (1 - x)$ changes sign once, so $J_s(z) > 0$ for all z once $J_s(1) > 0$, and $J_{17/8}(1) > 1/20000$ is an exact rational computation from $Q(x) \ge (1 - x^2 - x^{19}) \prod_{3 \le p \le 17} (1 - x^p)$ (script `moment_certificate.py`). Three rounds give $[1, 2] \to [1, 4] \to [1, 17/4]$. The method stops at $2\alpha_0 \approx 4.286$, where $\alpha_0 = \sup\{b : J_b(1) > 0\}$ (numerically; $J_3(1) < 0$ exactly). Some restriction on k is necessary (open problem 2).

5 The limiting value and its truncations

5.1 The limiting value

For $c_0 < 1$ the series $\sum_p c_0^p$ converges, so the product defining Q converges uniformly on $[0, c_0]$; hence Q is continuous on $[0, 1)$ and $\mathcal{L}$ is differentiable on $(0, 1)$ with $\mathcal{L}'(c) = Q(c)$. Write $I_N = F_1(\mathcal{P}_N)$, so that $I_N \downarrow \Lambda = \mathcal{L}(1)$. An exact rational sandwich gives

$$0.55065301127284296357797175 68\ldots < \Lambda < 0.55065301127284296357797175 71\ldots,$$

certifying 26 decimal places. The upper end is I_{2003}, computed exactly by integrating the integer coefficient vector of $\prod_{p\le 2003}(1 - x^p)$ (degree 279,053) termwise. For the lower end, every prime $p > 2003$ is odd and at least 2005, so $1 - \prod_{p>2003}(1 - x^p) \le x^{2005}/(1 - x^2)$ by the product inequality of Lemma 2.2, and we subtract $\int_0^1 x^{2005} \prod_{2<p\le 2003}(1 - x^p)\, dx$.

5.2 Tail asymptotics

Throughout, p runs over primes, $a = \zeta(2) = \pi^2/6$, $L(t) = \log(1/t)$, and for $u, t > 0$

$$S(u) = \sum_p e^{-pu}, \qquad W(u) = \sum_p \log p\, e^{-pu}, \qquad \Phi(t) = \sum_p -\log\bigl(1 - e^{-pt}\bigr),$$

so that $Q(e^{-t}) = e^{-\Phi(t)}$. Each summand of Φ is positive and decreasing in t, so Φ is decreasing. We use the prime number theorem in the form $\theta(x) = \sum_{p\le x} \log p = x + o(x)$.

Lemma 5.1 (Exponential prime sums)**.** *As $u \downarrow 0$, $uW(u) \to 1$ and $uL(u)S(u) \to 1$.*

Proof. Partial summation gives

$$W(u) = \sum_{n\ge 1} (\theta(n) - \theta(n-1)) e^{-nu} = (1 - e^{-u}) \sum_{n\ge 1} \theta(n) e^{-nu},$$

the series converging since $\theta(n) \le n \log 4$. Given $\varepsilon > 0$ there is C_ε with $|\theta(n) - n| \le \varepsilon n + C_\varepsilon$ for all n. Since $\sum_n n e^{-nu} = e^{-u}/(1 - e^{-u})^2$, we get

$$\left| W(u) - \frac{e^{-u}}{1 - e^{-u}} \right| \le \frac{\varepsilon e^{-u}}{1 - e^{-u}} + C_\varepsilon.$$

Multiplying by u and letting $u \downarrow 0$ gives $uW(u) \to 1$.

For S, fix $\delta \in (0, 1)$. Primes $p \le u^{-(1-\delta)}$ contribute at most $u^{-(1-\delta)} = o(1/(uL(u)))$ to $S(u)$. The remaining primes have $\log p \ge (1-\delta)L(u)$, so they contribute at most $W(u)/((1-\delta)L(u))$. Hence $\limsup uL(u)S(u) \le 1/(1 - \delta)$. Conversely, primes $p \le Y = u^{-(1+\delta)}$ have $\log p \le (1 + \delta)L(u)$, so

$$(1 + \delta)L(u)S(u) \;\ge\; W(u) - \sum_{n>Y} \log n\, e^{-nu} \;\ge\; W(u) - e^{-Yu/2} \sum_{n\ge 1} \log n\, e^{-nu/2}.$$

Here $Yu = u^{-\delta} \to \infty$ and the last sum is $O(u^{-2})$, so the subtracted term is $o(1/u)$ and $\liminf uL(u)S(u) \ge 1/(1 + \delta)$. Let $\delta \downarrow 0$. □

Lemma 5.2 (The prime partition exponent). *As $t \downarrow 0$, $tL(t)\Phi(t) \to \zeta(2)$.*

Proof. Expanding $-\log(1-y) = \sum_{m\ge1} y^m/m$ and exchanging the order of summation (all terms are nonnegative) gives $\Phi(t) = \sum_{m\ge1} S(mt)/m$. Since $L(t) = L(mt) + \log m$, the mth term of $tL(t)\Phi(t)$ is

$$\frac{tL(t)S(mt)}{m} = \frac{1}{m^2}\, mt\, L(mt)S(mt)\Big(1 + \frac{\log m}{L(mt)}\Big),$$

which tends to $1/m^2$ for each fixed m by Lemma 5.1. For domination, $S(u) \le \sum_{n\ge1} e^{-nu} \le 1/u$ for every $u > 0$, and by Lemma 5.1 there is $u_1 \in (0, e^{-1}]$ with $uL(u)S(u) \le 2$ for $0 < u < u_1$. If $mt < u_1$, then $L(mt) \ge 1$ and the mth term is at most $2(1+\log m)/m^2$. If $mt \ge u_1$, it is at most $L(t)/m^2 = (L(mt) + \log m)/m^2 \le (L(u_1) + \log m)/m^2$. The bound $(2 + L(u_1))(1 + \log m)/m^2$ is summable, so dominated convergence gives $tL(t)\Phi(t) \to \sum_m 1/m^2 = \zeta(2)$. □

Lemma 5.2 is the classical asymptotic for the logarithm of the generating function of partitions into primes; we include the short proof because the formalization follows it. The next lemma is unconditional. It compares $I_N - \Lambda$ with the function Φ, setting

$$M_N = \inf_{t>0}\big(Nt + \Phi(t)\big).$$

Lemma 5.3 (Sandwich). *(a) For every $N \ge 3$, $0 \le I_N - \Lambda \le e\, N^2 e^{-M_N}$.*

(b) Let $q > N$ be prime, $\tau > 0$ and $\varepsilon > 0$. If $\Phi(t) \le B$ for $\tau \le t \le (1+\varepsilon)\tau$, then

$$I_N - \Lambda \;\ge\; \frac{\varepsilon\tau e^{-\tau}}{1+\varepsilon\tau} \exp\big(-B - (1+\varepsilon)q\tau\big).$$

Proof. Write $P_N = P_{\mathcal{P}_N}$. For $M \ge N$ and $0 \le x < 1$,

$$0 \le P_N(x) - P_M(x) = P_N(x)\Big(1 - \prod_{N<p\le M}(1-x^p)\Big) \le P_N(x)\min\Big(1, \frac{x^{N+1}}{1-x}\Big),$$

by the product inequality of Lemma 2.2 and $\sum_{n>N} x^n = x^{N+1}/(1-x)$. Integrating and letting $M \to \infty$ ($I_M \to \Lambda$) gives $I_N - \Lambda \le \sup_{0<x<1} g(x)$, where $g(x)$ is the right side above. Write $x = e^{-t}$ with $t > 0$. For $y = x^n$ with $n > N$ one has $-\log(1-y) \le y/(1-y) \le x^n/(1-x)$. Summing over the primes $p > N$ gives

$$e^{-\Phi(t)} \;\le\; P_N(e^{-t}) \;\le\; \exp\Big(-\Phi(t) + \frac{e^{-(N+1)t}}{(1-e^{-t})^2}\Big). \tag{3}$$

Put $\ell = \log N \ge 1$ and $s = 2\ell/N$. If $t \ge s$, then $t \ge 2/N$ and $1 - e^{-t} \ge t/(1+t) \ge 2/(N+2) \ge 1/N$, while $e^{-(N+1)t} \le e^{-Nt} \le N^{-2}$. So the error term in (3) is at most 1, and

$$g(e^{-t}) \le e^{1-\Phi(t)} \cdot Ne^{-Nt} \le eNe^{-M_N}.$$

If $0 < t < s$, then $P_N(e^{-t}) \le P_N(e^{-s})$ because P_N is nonincreasing, so $g(e^{-t}) \le e^{1-\Phi(s)}$. Moreover $\Phi(s) \ge M_N - Ns = M_N - 2\ell$, hence $g(e^{-t}) \le eN^2e^{-M_N}$. This proves (a).

For (b), let $M \ge q$. Since $\mathcal{P}_N \cup \{q\} \subseteq \mathcal{P}_M$, $P_N - P_M \ge P_N - P_{\mathcal{P}_N\cup\{q\}} = x^q P_N \ge 0$ on $[0,1]$. On the window $x = e^{-t}$ with $\tau \le t \le (1+\varepsilon)\tau$, (3) gives $x^q P_N(x) \ge e^{-(1+\varepsilon)q\tau - B}$. The window has length $e^{-\tau}(1 - e^{-\varepsilon\tau}) \ge e^{-\tau}\varepsilon\tau/(1+\varepsilon\tau)$. Integrating and letting $M \to \infty$ proves (b). □

Proof of Theorem 1.5. Let $\ell = \log N$ and $\lambda_N = 2\sqrt{2aN/\ell} = 2\pi\sqrt{N/(3\ell)}$. Then $\ell = o(\lambda_N)$ and $\lambda_N = o(N)$. Fix $\varepsilon \in (0, \frac{1}{4})$. By Lemma 5.2 there is $t_0 \in (0,1)$ with

$$(1-\varepsilon)a \le tL(t)\Phi(t) \le (1+\varepsilon)a \qquad (0 < t < t_0).$$

Lower bound for $-\log(I_N - \Lambda)$. We bound $Nt + \Phi(t)$ from below for $t > 0$. If $t \ge t_0$, then $Nt + \Phi(t) \ge Nt_0$. If $t < t_0$ and $L(t) \le (\frac{1}{2} + \varepsilon)\ell$, then $\Phi(t) \ge b/t$ with $b = (1-\varepsilon)a/((\frac{1}{2}+\varepsilon)\ell)$, so

$$Nt + \Phi(t) \;\ge\; 2\sqrt{bN} = \sqrt{\frac{1-\varepsilon}{1+2\varepsilon}}\,\lambda_N.$$

Otherwise $t < t_1 = N^{-1/2-\varepsilon}$. Since Φ is decreasing, $\Phi(t) \ge \Phi(t_1) \ge (1-\varepsilon)aN^{1/2+\varepsilon}/((\frac{1}{2}+\varepsilon)\ell)$, provided $t_1 < t_0$. As Nt_0 and $N^{1/2+\varepsilon}/\ell$ are both $\gg \lambda_N$, we get $M_N \ge \sqrt{(1-\varepsilon)/(1+2\varepsilon)}\,\lambda_N$ for large N. Lemma 5.3(a) then gives

$$-\log(I_N - \Lambda) \;\ge\; M_N - 2\ell - 1 \;\ge\; \Big(\sqrt{\tfrac{1-\varepsilon}{1+2\varepsilon}} - o(1)\Big)\lambda_N.$$

Upper bound. By the prime number theorem, for large N there is a prime q with $N < q \le (1+\varepsilon)N$. Let $\tau = \sqrt{2a/(N\ell)}$, so that $N\tau = \lambda_N/2$ and $\tau \to 0$. For $\tau \le t \le (1+\varepsilon)\tau$ and large N, we have $t < t_0$ and

$$L(t) \;\ge\; L(\tau) - \log(1+\varepsilon) = \tfrac{1}{2}\big(\ell + \log\ell - \log 2a\big) - \log(1+\varepsilon) \;\ge\; (1-\varepsilon)\tfrac{\ell}{2}.$$

Hence $\Phi(t) \le (1+\varepsilon)a/(\tau(1-\varepsilon)\ell/2) = \frac{1+\varepsilon}{1-\varepsilon}N\tau$. By Lemma 5.3(b), with $(1+\varepsilon)q\tau \le (1+\varepsilon)^2 N\tau$ and $\log(1/\tau) \le \ell$,

$$-\log(I_N - \Lambda) \;\le\; \Big(\frac{1+\varepsilon}{1-\varepsilon} + (1+\varepsilon)^2\Big)\frac{\lambda_N}{2} + \ell + O_\varepsilon(1).$$

Both constants tend to 1 as $\varepsilon \downarrow 0$, which proves the first statement.

For the second, the prime number theorem gives $p_n \sim n\log n$, hence $\log p_n \sim \log n$ and $p_n/\log p_n \sim n$, so $\lambda_{p_n} \sim (2\pi/\sqrt{3})\sqrt{n}$. □

Remark 5.4 (Chebyshev bounds). With only $\theta(x) \asymp x$ in place of the prime number theorem, the same proof gives $-\log(I_N - \Lambda) \asymp \sqrt{N/\log N}$. Indeed, Lemmas 5.1 and 5.2 hold with $\asymp$ in place of limits, and Bertrand's postulate supplies $q \in (N, 2N]$. The prime number theorem is what fixes the constant $2\pi/\sqrt{3} = 2\sqrt{2\zeta(2)}$.

5.3 The next prime

Since $\mathcal{P}_N$ changes only at primes, $I_N = I_{q_N - 1}$, where q_N is the least prime exceeding N. The two halves of Lemma 5.3 are therefore sharpest when evaluated at q_N.

Proposition 5.5 (Next-prime sandwich)**.** *For every $N \ge 3$,*

$$M_{q_N} - 2\log q_N - 3 \;\le\; -\log(I_N - \Lambda) \;\le\; M_{q_N} + \log(q_N + 1) + 4.$$

Proof. Write $q = q_N \geq 5$. We first record $\Phi(1) < 1$: indeed $\Phi(1) \leq \sum_{n\geq 2} e^{-n}/(1-e^{-n}) \leq e^{-2}/((1-e^{-1})(1-e^{-2})) < 1$. Hence $M_q \leq q + \Phi(1) < q+1$.

Lower bound. Lemma 5.3(a) with $q-1 \geq 3$ in place of N gives $I_N - \Lambda = I_{q-1} - \Lambda \leq e(q-1)^2 e^{-M_{q-1}}$. Since $\Phi \geq 0$, $(q-1)t + \Phi(t) \geq (1-1/q)(qt + \Phi(t))$ for all $t > 0$, so $M_{q-1} \geq M_q - M_q/q > M_q - 2$. Therefore $-\log(I_N - \Lambda) \geq M_q - 2 - 1 - 2\log(q-1)$.

Upper bound. Choose $\tau > 0$ with $q\tau + \Phi(\tau) \leq M_q + 1$; then $q\tau < M_q + 1 < q + 2$, so $\tau < 2$. Apply Lemma 5.3(b) with this q and τ, with $\varepsilon = 1/(q\tau)$, and with $B = \Phi(\tau)$, which is admissible because Φ is decreasing. Since $\varepsilon\tau = 1/q$,

$$-\log(I_N - \Lambda) \leq \Phi(\tau) + q\tau + 1 + \tau + \log(q+1) \leq M_q + \log(q+1) + 4. \qquad \square$$

The proposition uses neither the prime number theorem nor any information about the size of $q_N - N$. To compare M_{q_N} with M_N we need the shape of M. The function Φ is C^1, strictly convex and strictly decreasing on $(0,\infty)$, since each summand $-\log(1-e^{-pt})$ is, and $\Phi(t) \to \infty$ as $t \downarrow 0$. Hence, for every real $N > 0$, $Nt + \Phi(t)$ has a unique minimizer t_N, characterized by $-\Phi'(t_N) = N$. As an infimum of affine functions of N, M is concave, and it has derivative $M'(N) = t_N$, which decreases in N. Consequently

$$(q-N)\,t_q \;\leq\; M_q - M_N \;\leq\; (q-N)\,t_N \qquad (N \leq q). \tag{4}$$

By convexity, $(\Phi(t) - \Phi(2t))/t \leq -\Phi'(t) \leq (\Phi(t/2) - \Phi(t))/(t/2)$, and Lemma 5.2 evaluates both sides. The result is $-\Phi'(t) \asymp 1/(t^2 L(t))$ as $t \downarrow 0$, with constants $a/2$ and $2a$ up to factors $1 + o(1)$. Since $-\Phi'(t_N) = N$, this gives $t_N \asymp (N \log N)^{-1/2}$.

Corollary 5.6 (Second-order term and prime gaps)**.** *The following are equivalent:*

(i) $-\log(I_N - \Lambda) \leq M_N + O(\log N)$ *for all* $N \geq 3$*;*

(ii) $M_{q_N} - M_N = O(\log N)$*;*

(iii) $p_{n+1} - p_n = O\big(\sqrt{p_n}\,(\log p_n)^{3/2}\big)$.

In particular they hold under the Riemann hypothesis. Unconditionally, $-\log(I_N - \Lambda) = M_N + O(N^{1/40})$.

Proof. By Bertrand's postulate $q_N \leq 2N$, so $\log q_N = O(\log N)$, and $M_{q_N} \geq M_N$. Then (i)⇒(ii) follows from the lower bound of Proposition 5.5, and (ii)⇒(i) from its upper bound. Since $t_q \asymp t_N \asymp (N\log N)^{-1/2}$ for $N \leq q \leq 2N$, (4) gives $M_{q_N} - M_N \asymp (q_N - N)(N\log N)^{-1/2}$. So (ii) is equivalent to $q_N - N = O(\sqrt{N}(\log N)^{3/2})$. This bound for all N is equivalent to (iii): take $N = p_n$ for one direction, and use $q_N - N \leq p_{n+1} - p_n$ for $p_n \leq N < p_{n+1}$ for the other. Under the Riemann hypothesis Cramér [5] proved $p_{n+1} - p_n = O(\sqrt{p_n}\log p_n)$. Unconditionally, Baker, Harman and Pintz [2] proved $p_{n+1} - p_n = O(p_n^{0.525})$, so $M_{q_N} - M_N = O(N^{0.525}(N\log N)^{-1/2}) = O(N^{1/40})$; combine this with Proposition 5.5. $\square$

Thus refining Theorem 1.5 to additive precision $O(\log N)$ is a statement about gaps between primes, and it is out of reach unconditionally. The truncation error sees the gap after N directly. For instance, at the record gap $31397 \to 31469$, I_N stays constant for 72 consecutive values of N while M_N increases.

5.4 Stability of the minimizer

How much must F_c increase when a primitive set omits a prime? The relative theorem answers this exactly.

Proposition 5.7 (Omitting a prime)**.** *Let p be prime and $B_p = (\mathcal{P} \setminus \{p\}) \cup \{p^2\}$. For every $c \in (0,1]$, B_p is the unique minimizer of F_c among primitive sets A with $p \notin A$, and*

$$F_c(B_p) = \mathcal{L}(c) + D_c(p), \qquad D_c(p) = \sum_{j=1}^{p-1} \int_0^c x^{jp} Q(x)\,dx \;\geq\; \Delta_c(p) := \int_0^c x^p Q(x)\,dx > 0,$$

with $D_c(2) = \Delta_c(2)$.

Proof. A primitive set avoids p exactly when it lies in $U = \{2,3,\dots\}\setminus\{p\}$, and $\mathrm{Min}(U) = B_p$: a composite with a prime factor $q \neq p$ has the proper divisor $q \in U$; p^k with $k \geq 3$ has the proper divisor $p^2 \in U$; primes $q \neq p$ and p^2 have no proper divisor in U. Theorem 4.1 gives the first statement. On $[0,1)$, $Q = (1-x^p)P_{\mathcal{P}\setminus\{p\}}$ and $P_{B_p} = (1-x^{p^2})P_{\mathcal{P}\setminus\{p\}}$, so $P_{B_p} = Q\cdot(1-x^{p^2})/(1-x^p) = Q\sum_{j=0}^{p-1} x^{jp}$; integrate. □

Corollary 5.8 (Stability)**.** *Let A and $A_1, A_2, \dots$ be primitive sets.*

(a) If $p \notin A$ then $F_c(A) - \mathcal{L}(c) \geq D_c(p) \geq \Delta_c(p)$ for every $c \in (0,1]$.

(b) If $F_1(A) - \Lambda < \Delta_1(p)$ for every prime $p \leq R$, then $A \cap [2,R] = \mathcal{P}_R$. Since Δ_1 is decreasing in p, it suffices that $F_1(A) - \Lambda < \Delta_1(r)$ for the largest prime $r \leq R$.

(c) $F_1(A_j) \to \Lambda$ if and only if every prime belongs to A_j for all large j.

Proof. (a) is Proposition 5.7. For (b), (a) shows that every prime $p \leq R$ lies in A, and then no composite $m \leq R$ lies in A, since its prime factors are at most R. For (c), if some prime is omitted infinitely often then (a) bounds $F_1(A_j) - \Lambda$ below along a subsequence. Conversely, if every prime is eventually in A_j, then by primitivity every fixed composite is eventually excluded; the indicator of A_j converges pointwise to that of $\mathcal{P}$, so $P_{A_j} \to Q$ on $[0,1)$ by dominated convergence for the products, and $F_1(A_j) \to \Lambda$ by dominated convergence for the integrals. □

The threshold $\Delta_1(p)$ is governed by the same saddle as the truncation error.

Proposition 5.9 (Size of the threshold)**.** *For every prime, and indeed every integer, $p \geq 2$,*

$$M_p \;\leq\; -\log\Delta_1(p) \;\leq\; M_p + \log p + 3,$$

and consequently $-\log\Delta_1(p) \sim 2\pi\sqrt{p/(3\log p)}$.

Proof. With $x = e^{-t}$, $x^p Q(x) = e^{-(pt+\Phi(t))} \leq e^{-M_p}$, which gives the lower bound after integrating over $(0,1)$. For the upper bound, fix $\tau > 0$ and integrate over the window $t \in [\tau, \tau + 1/p]$, where $pt + \Phi(t) \leq p\tau + \Phi(\tau) + 1$ because Φ is decreasing. The window has x-length $e^{-\tau}(1 - e^{-1/p}) \geq e^{-\tau}/(p+1)$, so

$$-\log\Delta_1(p) \leq p\tau + \Phi(\tau) + \tau + 1 + \log(p+1) \leq p\tau + \Phi(\tau) + \tau + \frac{1}{p} + 1 + \log p.$$

Choose τ with $p\tau + \Phi(\tau) \le M_p + \delta$, where $\delta = (1 - \Phi(1))/3 > 0$ (recall $\Phi(1) < 1$ from the proof of Proposition 5.5). Since $M_p \le p + \Phi(1)$, this gives $\tau \le 1 + (\Phi(1) + \delta)/p$, and the bound becomes $M_p + \log p + 2 + \delta + (\Phi(1) + \delta + 1)/p \le M_p + \log p + 3$ for $p \ge 2$. For the asymptotic, the proof of Theorem 1.5 gives $M_p \ge (1 - o(1))\lambda_p$. Lemma 5.3(a) gives $M_p \le -\log(I_p - \Lambda) + 2\log p + 1$, and Theorem 1.5 gives $-\log(I_p - \Lambda) \sim \lambda_p$. □

So a primitive set whose F_1 lies within $e^{-(1+o(1))2\pi\sqrt{R/(3\log R)}}$ of Λ agrees with the primes up to R (Corollary 5.8(b) with the largest prime $r \le R$, and $r \sim R$). This scale is sharp: the truncation $\mathcal{P}_{r-1}$ omits r, and its excess is $I_{r-1} - \Lambda = e^{-(1+o(1))\lambda_r}$ by Theorem 1.5.

5.5 Second-order terms

Relative refinements of Theorem 1.5 do not need short gaps. Put $a = \zeta(2)$, $\ell = \log N$, $\lambda_N = 2\sqrt{2aN/\ell} = 2\pi\sqrt{N/(3\ell)}$,

$$\sigma = \sum_{m\ge1} \frac{\log m}{m^2} = -\zeta'(2), \qquad \kappa = \gamma + \frac{\sigma}{a}, \qquad C = \kappa + \tfrac{1}{2}\log(2a) = 1.7426003995\ldots,$$

where γ is Euler's constant.

Theorem 5.10 (Second-order tail asymptotics)**.** *As $N \to \infty$,*

$$-\log(I_N - \Lambda) = \lambda_N\Big(1 - \frac{\log\ell}{2\ell} + \frac{C}{\ell} + O\Big(\frac{(\log\ell)^2}{\ell^2}\Big)\Big).$$

The only arithmetic input is the prime number theorem with the error term $\theta(x) = x + O(x/\log^2 x)$, which follows from the classical de la Vallée Poussin error term. The proof has four steps.

Step 1: prime sums. Put $L = \log(1/u)$. Then

$$S(u) = \frac{1}{uL}\Big(1 + \frac{\gamma}{L}\Big) + O\Big(\frac{1}{uL^3}\Big) \qquad (u \downarrow 0).$$

Partial summation gives $S(u) = \sum_{n\ge2} \theta(n)\big(g(n) - g(n+1)\big)$ with $g(x) = e^{-xu}/\log x$. The error term of θ contributes $O(\sum_n (e^{-nu}/\log^4 n + nu\, e^{-nu}/\log^3 n)) = O(1/(uL^3))$. The main term equals $\sum_{n\ge2} g(n) + O(1) = \int_2^\infty g + O(1)$. Finally, $u\int_2^\infty g = \int_{2u}^\infty e^{-y}\,dy/(L + \log y)$; expanding $1/(L + \log y)$ to second order and using $\int_0^\infty e^{-y}\log y\,dy = \Gamma'(1) = -\gamma$ gives the claim.

Step 2: the exponent. Put $L = \log(1/t)$. Then

$$\Phi(t) = \frac{a}{tL}\Big(1 + \frac{\kappa}{L}\Big) + O\Big(\frac{1}{tL^3}\Big) \qquad (t \downarrow 0).$$

Indeed $\Phi(t) = \sum_m S(mt)/m$ (proof of Lemma 5.2). The terms with $m > t^{-1/2}$ contribute $O(t^{-1/2})$, because $S(u) \le 1/u$. For $m \le t^{-1/2}$, Step 1 applies with $L - \log m \ge L/2$ in place of L, and expanding $1/(L - \log m)$ in powers of $1/L$ gives $\sum_m m^{-2}\big(1/L + (\gamma + \log m)/L^2\big)/t$ up to $O(1/(tL^3))$. Here $\sum m^{-2} = a$ and $\sum m^{-2}\log m = \sigma$.

Step 3: the saddle. Let $u_0 = \frac{1}{2}(\ell + \log\ell - \log 2a)$, so $e^{2u_0} = N\ell/(2a)$, and $\tau = e^{-u_0}$; then $N\tau = \lambda_N/2$. Put

$$S_N = 1 - \frac{\log\ell}{2\ell} + \frac{C}{\ell}, \qquad r_N = \frac{\log\ell}{\ell}.$$

Since Φ is nonnegative and nonincreasing, Step 2 and Lemma A.1 (Appendix A) give a constant K such that, for all large N,

$$\lambda_N\big(S_N - Kr_N^2\big) \le Nt + \Phi(t) \quad (t>0), \qquad N\tau + \Phi(\tau) \le \lambda_N\big(S_N + Kr_N^2\big).$$

In particular $M_N = \inf_{t>0}(Nt+\Phi(t)) = \lambda_N\big(S_N + O(r_N^2)\big)$.

Step 4: transfer. Lemma 5.3(a) gives $-\log(I_N - \Lambda) \ge M_N - 2\log N - 1$. For the upper bound, the error term of θ provides a prime $q \in (N, N + O(N/\ell^2)]$. Apply Lemma 5.3(b) with $\tau = e^{-u_0}$, $\varepsilon = \ell^{-2}$ and $B = \Phi(\tau)$, which is admissible because Φ is decreasing:

$$-\log(I_N - \Lambda) \le N\tau + \Phi(\tau) + (q(1+\varepsilon) - N)\tau + O(\log N),$$

and $(q(1+\varepsilon)-N)\tau = O(N\tau/\ell^2) = O(\lambda_N/\ell^2)$. Since $\log N = o(\lambda_N(\log\ell/\ell)^2)$, the theorem follows.

Remark 5.11 (Partitions into primes). Let $p_{\mathcal{P}}(N)$ be the number of partitions of N into primes, so that $\sum_N p_{\mathcal{P}}(N)x^N = 1/Q(x) = e^{\Phi(t)}$ with $x = e^{-t}$. By Roth and Szekeres [15], $\log p_{\mathcal{P}}(N) \sim 2\pi\sqrt{N/(3\log N)}$, so Theorem 1.5 says $-\log(I_N - \Lambda) \sim \log p_{\mathcal{P}}(N)$: both are governed by the saddle point of $Nt + \Phi(t)$, although the proofs above do not use partitions. More precisely,

$$\log p_{\mathcal{P}}(N) = M_N + O(\log N). \tag{5}$$

The upper bound $\log p_{\mathcal{P}}(N) \le M_N$ holds for every N, since $p_{\mathcal{P}}(N)e^{-Nt} \le e^{\Phi(t)}$. For the lower bound, Vaughan's asymptotic formula [18], in the saddle-point form of Gafni [6, Theorem 1.3 with $k=1$], gives $p_{\mathcal{P}}(N) = e^{M_N}\big(2\pi\Phi''(t_N)\big)^{-1/2}\big(1 + O(N^{-1/5})\big)$, with t_N as in Section 5.3, and $\log\Phi''(t_N) = O(\log N)$: the term $p=2$ gives $-\Phi'(t) \ge 1/t - 1$, so $t_N \ge 1/(N+1)$, while $Nt_N \le M_N \le N + \Phi(1) < N+1$ (proof of Proposition 5.5) gives $t_N \le 2$; on $(0,2]$, $4e^{-4} \le \Phi''(t) \le 2(1+t)^5 t^{-5}$, using $1 - e^{-pt} \ge t/(1+t)$ and $\sum_n n^2 y^n \le 2/(1-y)^3$. (The prefactor contributes $-\frac{3}{4}\log N + O(\log\log N)$ [6, (1.5)].) By (5) and Corollary 5.6, the difference $-\log(I_N - \Lambda) - \log p_{\mathcal{P}}(N)$ is governed by the gap after N. Since $\log N = o(\lambda_N r_N^2)$, Step 3 above also gives $\log p_{\mathcal{P}}(N) = \lambda_N\big(S_N + O(r_N^2)\big)$. This agrees with the formal saddle-point expansion of Bartel, Bhaduri, Brack and Murthy [4, (43)]: their coefficient of $\log\ell/\ell$ is $-\frac{1}{2}$ (which they give in place of the coefficient $+1$ in [18]), and their coefficient of $1/\ell$ is $f_2/f_1 + \log(\pi/\sqrt{3})$ with $f_1 = a$ and $f_2 = \gamma a + \sigma$ [4, (23)], which equals $\kappa + \frac{1}{2}\log(2a) = C$.

5.6 Open problems

1. Is there a closed form for Λ, as for the all-exponent and even members of the family? Is Λ irrational?

2. Characterize the increasing functions ϕ for which $\mathcal{P}_N$ minimizes $\mathbb{E}\phi(Z_A)$ over primitive $A \subseteq \{2,\dots,N\}$ for every N (notation of Remark 4.4). This convex cone contains the increasing concave functions and the powers x^k, $0 < k \le 17/4$, and excludes the indicators $\mathbf{1}_{(x,1]}$ for x near 1 (Proposition 1.3). It omits x^k for large k: for finite A, $\mathbb{E}Z_A^k \sim \big(\prod_{a\in A} a\big)|A|!\,k^{-|A|}$ as $k\to\infty$, and already $\mathbb{E}Z_{A_9}^k < \mathbb{E}Z_{\mathcal{P}_9}^k$ for every $k \ge 126$ (exact computation, `powers_check.py`). The bound $2\alpha_0 \approx 4.286$ of Remark 4.4 is not sharp: for $N \in \{12, 18, 24\}$ and every maximal primitive subset of $\{2,\dots,N\}$, the primes give the smaller $\mathbb{E}\min(Z^k, c^k)$ for eleven exponents $k \in [5/2, 20]$ and $c \in \{\frac{1}{2}, \frac{9}{10}, \frac{19}{20}, \frac{99}{100}, 1\}$ (`moments_search.py`; 40-digit arithmetic, not a proof). What is the largest admissible k?

3. Proposition 5.5 determines $-\log(I_N - \Lambda)$ up to $O(\log N)$ by M_{q_N}, and Corollary 5.6 shows that replacing M_{q_N} by M_N at this precision is a prime-gap problem. What is the next term relative to M_{q_N}? Writing $I_N - \Lambda$ as the sum over primes $p > N$ of $\int_0^1 x^p P_{\mathcal{P}_{<p}}(x)\,dx$, applying Laplace's method to each term, and giving the primes near q_N their average density on the scale $\sqrt{N \log N}$ suggests

$$-\log(I_N - \Lambda) = M_{q_N} + \tfrac{1}{4}\log q_N + \tfrac{3}{4}\log\log q_N - \tfrac{1}{2}\log\pi + \tfrac{1}{4}\log(2\zeta(2)) + o(1).$$

Numerically, at $N = p_n$ for $20 \le n \le 9000$ ($N \le 93{,}000$, including the maximal gaps after 19609 and 31397), the left side minus the right side, without the $o(1)$, lies between -0.23 and 0.11 (script `next_prime_check.py`; floating point, not certified). A proof for every N would need control of primes in intervals of length about $\sqrt{N}$ after N, beyond current unconditional methods. Does the formula hold for almost all N?

6 Formal verification

The accompanying Lean 4/Mathlib development [7, 10] (`https://github.com/alerad/prime-extremality`, release `v1.0.1`, commit `f6caaa43`, [13]) is built with Lean `v4.35.0`, Mathlib commit `56a005855ea17e0b0958f73ec801ae140b0a1679`, the LeanCert library (Bernstein certificates, interval arithmetic and exact q-product integrals), and the prime number theorem imported, unmodified, from the PrimeNumberTheoremAnd project [12] at commit `127b50e5`. It formalizes, for every $c \in (0, 1]$ and for finite and infinite sets: Theorems 1.1, 1.2, 1.4 (by the route of Section 2) and 4.1; Propositions 1.3 and 3.4; the increasing-concave minimality of Remark 4.4 for finite sets and its moment extension for all real $0 < k \le 17/4$, including the exact values of $J_{17/8}(1)$ and $J_3(1)$; the 26 digits of Λ; Lemmas 5.1, 5.2, 5.3 and A.1; Theorems 1.5 and 5.10; Proposition 5.7, Corollary 5.8 and Proposition 5.9; and the comparison at $k = 126$ in open problem 2. The prime number theorem enters through `chebyshev_asymptotic` and `nth_prime_asymp` for Theorem 1.5, and through `MediumPNT` together with Mathlib's bound for $\psi - \theta$ for the hypothesis $\theta(x) = x + O(x/\log^2 x)$ of Theorem 5.10. Not formalized are the probabilistic model of Remark 4.4 itself, the infinite case of its increasing-concave statement, the numerical value of α_0, the identification of $\mathrm{Min}(U)$ in Corollary 4.2, Proposition 5.5, Corollary 5.6, and Remarks 5.4, 5.11 and 4.3. Some Lean proofs take equivalent routes (for instance dominated convergence in place of Stieltjes integration by parts); the repository lists each deviation and maps every result to its Lean declaration. An earlier artifact [14] verified a Bernstein-certificate proof of Theorem 1.4.

All of these results depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`, except the digits of Λ and the comparison in open problem 2, whose finite rational computations use `native_decide` and hence trust Lean's compiler.

A The saddle estimate

This appendix proves the estimate used in Step 3 of the proof of Theorem 5.10. The lemma is stated for an abstract function Φ; its Lean counterpart is `saddle_second_order` in `Tail/Saddle2.lean`, and the proof below follows it.

Lemma A.1 (Second-order saddle estimate)**.** *Let $a > 0$, $\kappa \in \mathbb{R}$, $K_1 \geq 0$ and $t_0 \in (0,1)$, and let $\Phi : (0,\infty) \to [0,\infty)$ be nonincreasing with*

$$\left|\Phi(t) - \frac{a}{tL}\Big(1 + \frac{\kappa}{L}\Big)\right| \leq \frac{K_1}{tL^3} \qquad (0 < t < t_0,\ L = \log(1/t)). \tag{6}$$

For $N > e$ put $\ell = \log N$, $\lambda_N = 2\sqrt{2aN/\ell}$, $u_0 = \frac{1}{2}(\ell + \log\ell - \log 2a)$, $\tau = e^{-u_0}$, $C = \kappa + \frac{1}{2}\log(2a)$, $S_N = 1 - \log\ell/(2\ell) + C/\ell$ and $r_N = \log\ell/\ell$. There is a constant K, depending only on a, κ, K_1, such that for all sufficiently large N:

(i) $Nt + \Phi(t) \geq \lambda_N(S_N - Kr_N^2)$ for every $t > 0$;

(ii) $N\tau + \Phi(\tau) \leq \lambda_N(S_N + Kr_N^2)$, and $N\tau = \lambda_N/2$.

Proof. Notation. Write $K_0 = K_1/a$ and $h_\pm(u) = 1/u + \kappa/u^2 \pm K_0/u^3$. In the variable $u = \log(1/t)$, (6) says

$$ae^u h_-(u) \leq \Phi(e^{-u}) \leq ae^u h_+(u) \qquad (u > \log(1/t_0)). \tag{7}$$

Put $X = e^{u_0}$ and $A = N/X$. Then $X^2 = N\ell/(2a)$, so $A = 2aX/\ell$, $A^2 = 2aN/\ell$ and $\lambda_N = 2A$. For $t = e^{-(u_0+v)}$,

$$Nt = Ae^{-v}, \qquad ae^{u_0+v} = A\,\tfrac{\ell}{2}\,e^v. \tag{8}$$

Let $r = r_N$, $D = 1 + |\log 2a|$, $x = (2u_0 - \ell)/\ell = (\log\ell - \log 2a)/\ell$ and $p = \ell/(2u_0) = 1/(1+x)$. Then $S_N = 1 - x/2 + \kappa/\ell$. We take N so large that $\log\ell \geq 1$ and

$$(D + |\kappa| + 2)\,r \leq \tfrac{1}{40}, \qquad u_0 \geq U := 20(|\kappa| + K_0) + 1, \qquad Xt_1 \geq 4, \tag{9}$$

where $t_1 = \min(t_0, e^{-U})$. Then $1/\ell \leq r \leq 1/80$, so $\ell \geq 80$; $|x| \leq Dr \leq 1/40$, so $p \leq 2$; and $S_N \leq \frac{6}{5}$.

Algebra at u_0. Put $P_\pm = \frac{\ell}{2}h_\pm(u_0) = p + 2\kappa p^2/\ell \pm 4K_0p^3/\ell^2$. Using $p = 1 - x + x^2p$ and $p^2 - 1 = -xp(p+1)$,

$$1 + P_\pm - 2S_N = x^2p + \frac{2\kappa(p^2-1)}{\ell} \pm \frac{4K_0p^3}{\ell^2},$$

so that

$$|1 + P_\pm - 2S_N| \leq K_2r^2, \qquad K_2 = 2D^2 + 12|\kappa|D + 32K_0. \tag{10}$$

Similarly $\delta = P_- - 1 = (p-1) + 2\kappa p^2/\ell - 4K_0p^3/\ell^2$ satisfies $|\delta| \leq K_3 r$ with $K_3 = 2D + 8|\kappa| + 32K_0$, because $|p-1| = |x|p$.

Two elementary inequalities. First, $e^{-v} + e^v \geq 2 + v^2$ for all real v, and $|e^v - 1| \leq 2|v|$ for $v \leq 1$ (for $0 \leq v \leq 1$, $e^v - 1 \leq v(e-1)$; for $v < 0$, $1 - e^v \leq \min(1, |v|)$). Hence for $v \leq 1$, $\delta \in \mathbb{R}$ and $B \geq 0$,

$$e^{-v} + (1+\delta)e^v - B\max(v,0) \geq 2 + \delta + \tfrac{v^2}{2} - (2|\delta| + B)|v| \geq 2 + \delta - \tfrac{1}{2}(2|\delta| + B)^2. \tag{11}$$

Second, h_- is nonincreasing on $[U,\infty)$, since $h_-'(u) = -u^{-2}(1 + 2\kappa/u - 3K_0/u^2)$ and $(2|\kappa| + 3K_0)/u < 1$ there. Moreover, for $u_0 > 0$ and $0 \leq v \leq 1$,

$$h_-(u_0 + v) \geq h_-(u_0) - \frac{v}{u_0^2} - \frac{2|\kappa|}{u_0^3}, \tag{12}$$

because $1/(u_0+v) \ge 1/u_0 - v/u_0^2$, $0 \le u_0^{-2} - (u_0+v)^{-2} \le 2u_0^{-3}$ and $(u_0+v)^{-3} \le u_0^{-3}$.

Proof of (i). Fix $t > 0$.

Case $t \ge t_1$. Since $\Phi \ge 0$, $Nt + \Phi(t) \ge Nt_1 = AXt_1 \ge 4A = 2\lambda_N \ge \lambda_N S_N$.

Case $t < t_1$. Write $t = e^{-(u_0+v)}$; then $u_0 + v > U$ and $t < t_0$, so (7) and (8) give

$$Nt + \Phi(t) \ge A\Big(e^{-v} + \tfrac{\ell}{2} e^{v} h_-(u_0+v)\Big).$$

If $v \le 0$, then $U < u_0 + v \le u_0$ and monotonicity of h_- gives $\frac{\ell}{2} h_-(u_0+v) \ge P_-$, so by (11) with $B = 0$, $Nt + \Phi(t) \ge A(2 + \delta - 2\delta^2)$. If $0 < v \le 1$, then (12) and $e^v \le 3$ give $\frac{\ell}{2} e^v h_-(u_0+v) \ge P_- e^v - Bv - E$ with $B = 3\ell/(2u_0^2) = 6p^2/\ell \le 24r$ and $E = 3|\kappa|\ell/u_0^3 = 24|\kappa|p^3/\ell^2 \le 192|\kappa|r^2$; by (11), $Nt + \Phi(t) \ge A\big(2 + \delta - \frac{1}{2}(2|\delta| + B)^2 - E\big)$. In both subcases, by (10) and $2 + \delta = 1 + P_-$,

$$Nt + \Phi(t) \ge A\Big(2S_N - K_2 r^2 - \tfrac{1}{2}(2K_3 + 24)^2 r^2 - 192|\kappa| r^2\Big) = \lambda_N (S_N - K r^2),$$

with $K = \frac{1}{2}K_2 + \frac{1}{4}(2K_3 + 24)^2 + 96|\kappa|$. Finally, if $v > 1$, put $t_2 = e^{-(u_0+1)} < t_1$. Since Φ is nonincreasing, (7) and (8) give

$$\Phi(t) \ge \Phi(t_2) \ge Ae\, \tfrac{\ell}{2} h_-(u_0+1) = Ae\, \frac{\ell}{2(u_0+1)}\big(1 + \kappa q - K_0 q^2\big), \qquad q = \frac{1}{u_0+1}.$$

Here $\ell/(2(u_0+1)) = \ell/(\ell(1+x)+2) \ge \frac{19}{20}$ because $x \le \frac{1}{40}$ and $\ell \ge 80$, and $1 + \kappa q - K_0 q^2 \ge 1 - (|\kappa| + K_0) q \ge \frac{19}{20}$ because $u_0 + 1 \ge U$. Hence $\Phi(t) \ge (\frac{19}{20})^2 eA > 2.4A \ge \lambda_N S_N$. This proves (i), since $S_N \ge S_N - Kr^2$.

Proof of (ii). By (9), $\tau = 1/X \le t_1/4 < t_0$, and $N\tau = A = \lambda_N/2$. By (7) and (8) with $v = 0$, $\Phi(\tau) \le aXh_+(u_0) = AP_+$, so by (10)

$$N\tau + \Phi(\tau) \le A(1 + P_+) \le A(2S_N + K_2 r^2) \le \lambda_N (S_N + K r^2). \qquad \square$$

Declarations

Use of AI. The author used Claude Code (Anthropic) in developing parts of the Lean code, the verification scripts, and the manuscript. The author checked the mathematical arguments and assumes responsibility for the paper.

Competing interests. The author declares no competing interests.

Data availability. The Lean development of Section 6 and the supplementary scripts `crossing_endpoint.py`, `powers_check.py`, `moment_certificate.py`, `moments_search.py`, `tail_asymptotic_check.py` and `next_prime_check.py` are publicly available at https://github.com/alerad/prime-extremality and archived at https://doi.org/10.5281/zenodo.23537236 [13].

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